5.1-5.4 Advanced Security

Miscellaneous Security Settings

Only for the cool intermediate kids

brought to you by clem, ceo of subway



Run: “control system” or Win+Pause|Break

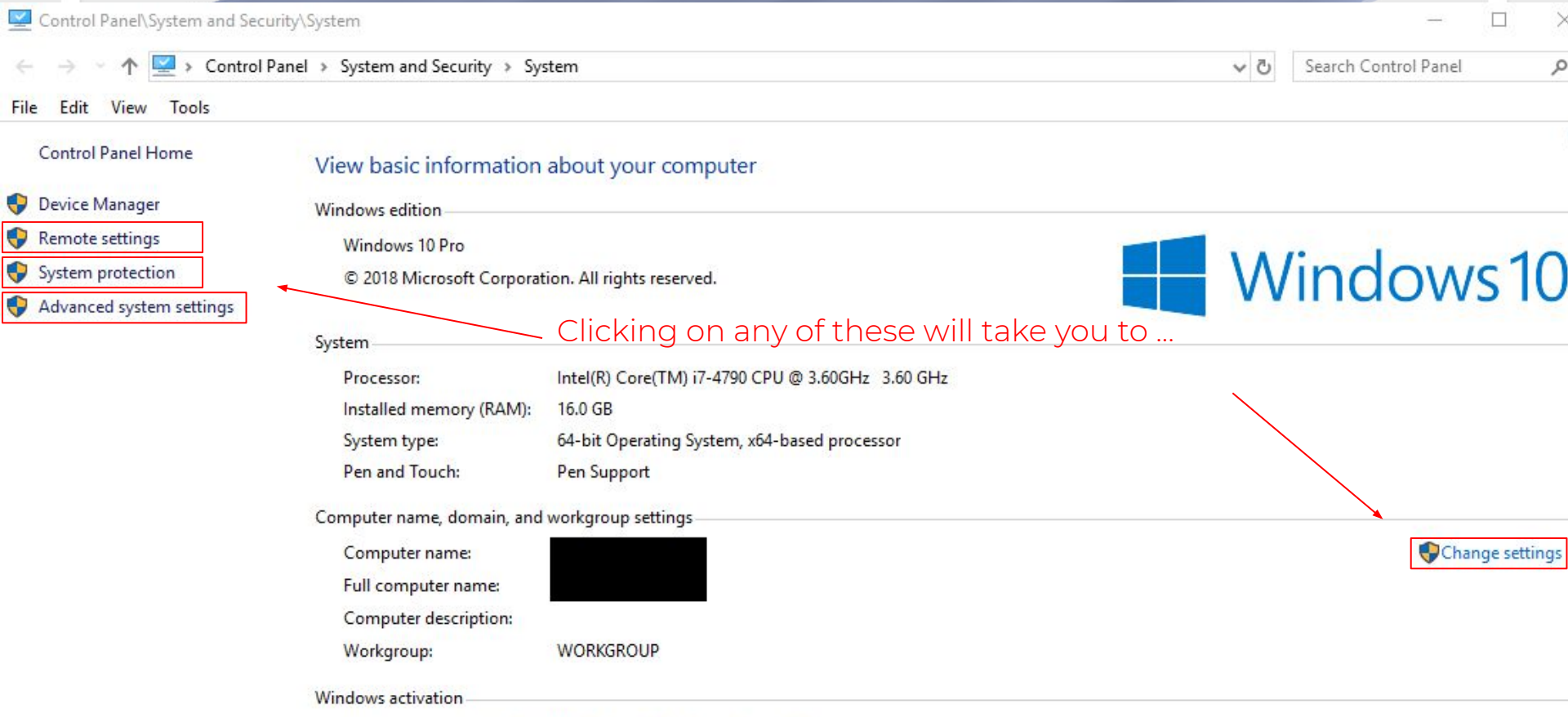
System info

- Edition
- Processor
- RAM

Network info

- Computer name
- Domain/workgroup

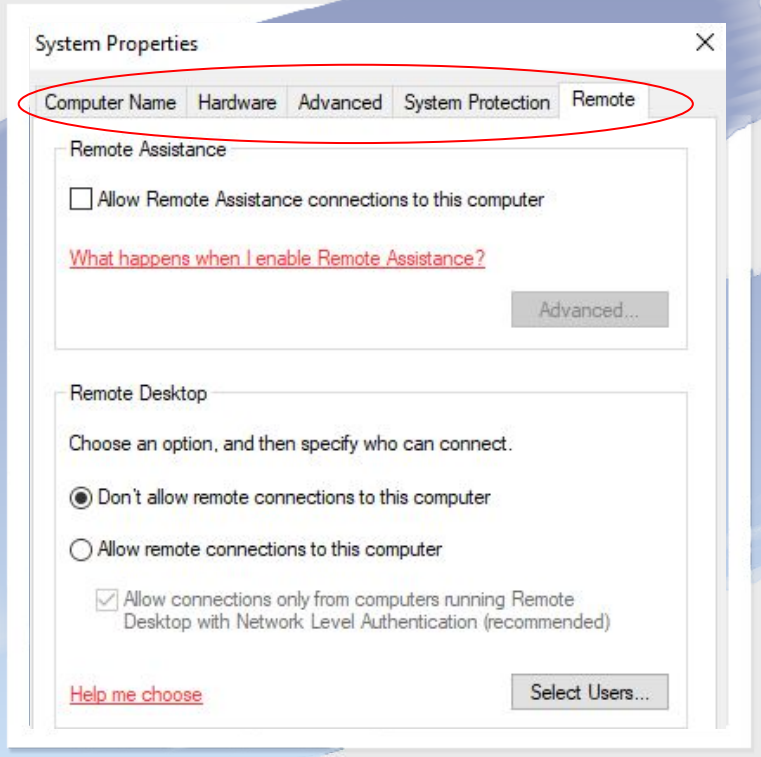
SYSTEM


Control Panel\System and Security\System
Control Panel › System and Security › System
Search Control Panel
File Edit View Tools
Control Panel Home
Device Manager
Remote settings
System protection
Advanced system settings
View basic information about your computer
Windows edition
Windows 10 Pro
© 2018 Microsoft Corporation. All rights reserved.
Windows 10
Clicking on any of these will take you to ...
System
Processor: Intel(R) Core(TM) i7-4790 CPU @ 3.60GHz 3.60 GHz
Installed memory (RAM): 16.0 GB
System type: 64-bit Operating System, x64-based processor
Pen and Touch: Pen Support
Computer name, domain, and workgroup settings
Computer name:
Full computer name:
Computer description:
Workgroup: WORKGROUP
Change settings
Windows activation

5.1 System Properties

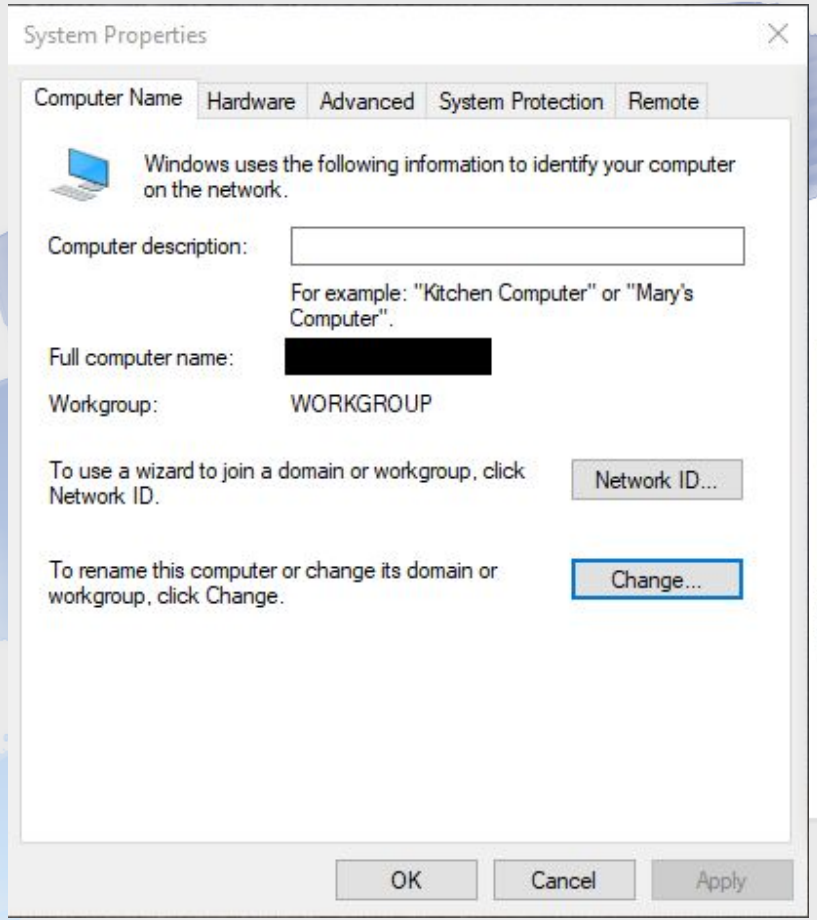
System Properties

There are more tabs than “Remote”



Computer Name

- Change description
- Change network ID
  - Joining workgroup/domain
- Change computer name/domain



Computer Groups

- **Homegroup** - network of computers in a home
- **Workgroup** - network of computers in a workplace
- **Domain** - network of computers in a corporation

Homegroups and workgroups are decentralized

No central computer handling all the users and computers etc

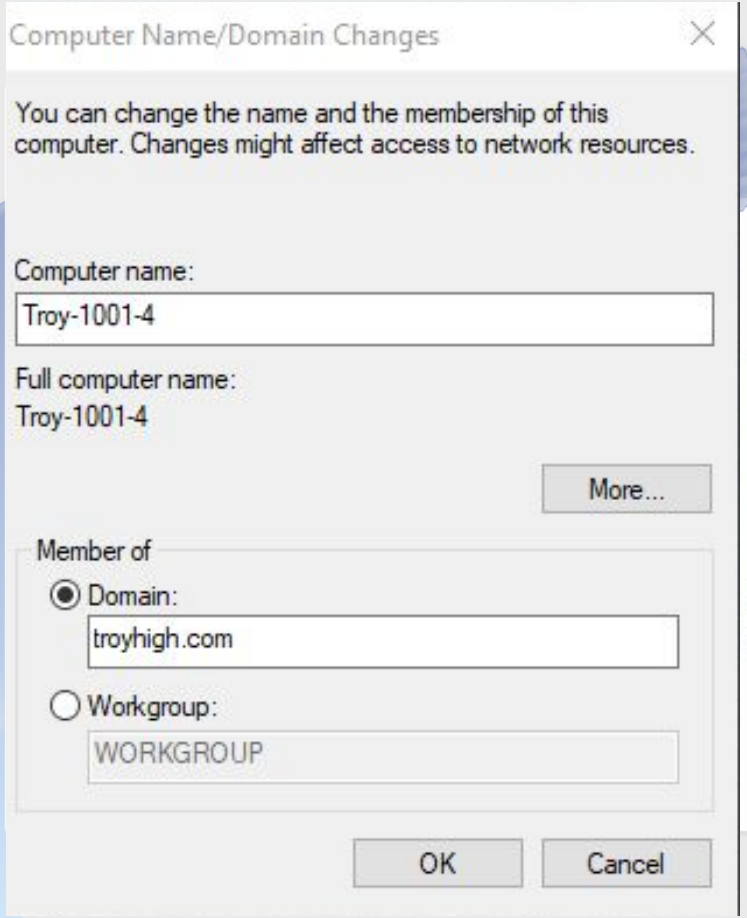
Why Rename?

Organization

Instead of “Please fix DESKTOP-DNM5QRM”

“Please fix Troy-1001-4”

- as in “the 4th computer in room 1001 at Troy”



Why Rename? (cont.)

Quick access in a private network

WIN-5LATHCLO73A.contoso.com

vs

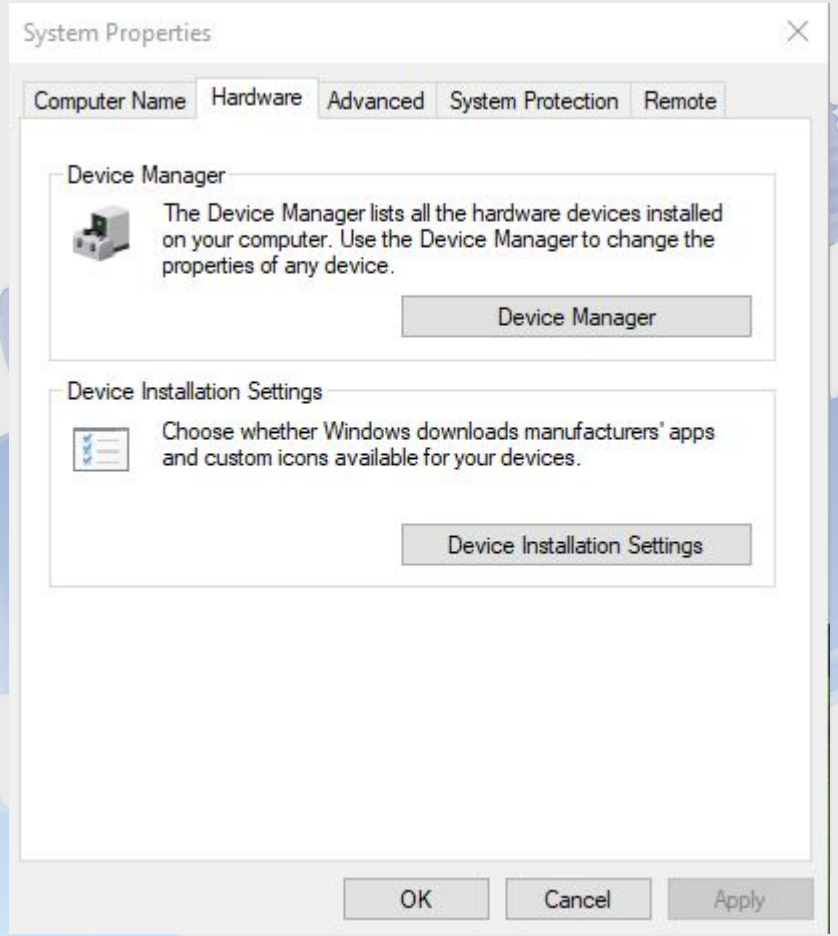
mail.contoso.com

DNS does make this easier without needing to rename your computers, however (will talk about later)

Hardware

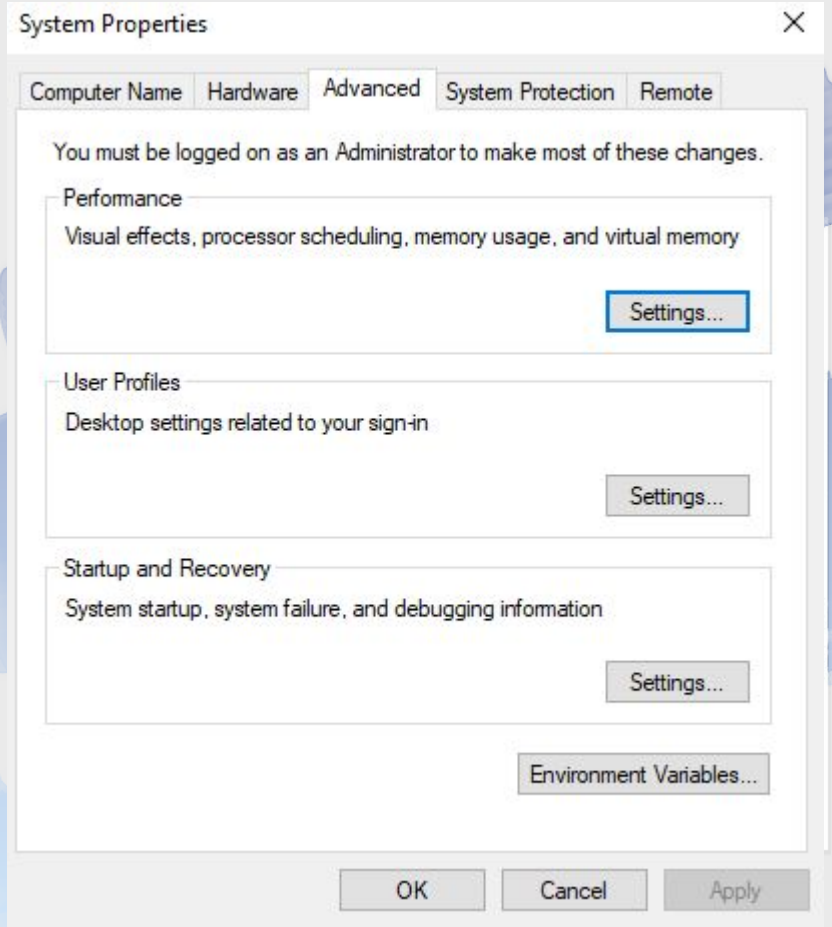
- Access the device manager
- Change settings for driver updates

More for tech support than security



Advanced

- Change performance
- Handle user profiles
- Startup settings
- Environment variables (useful for commands and others)

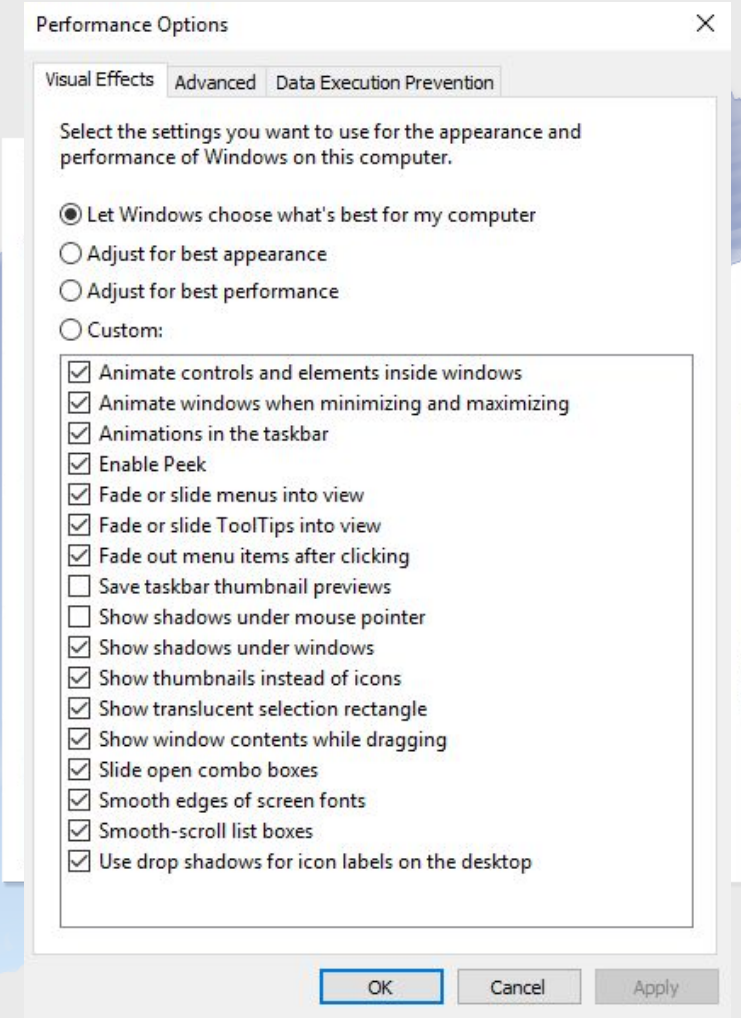


Performance Options

Make your VM go “i am speed”

Adjust for best performance if you wish

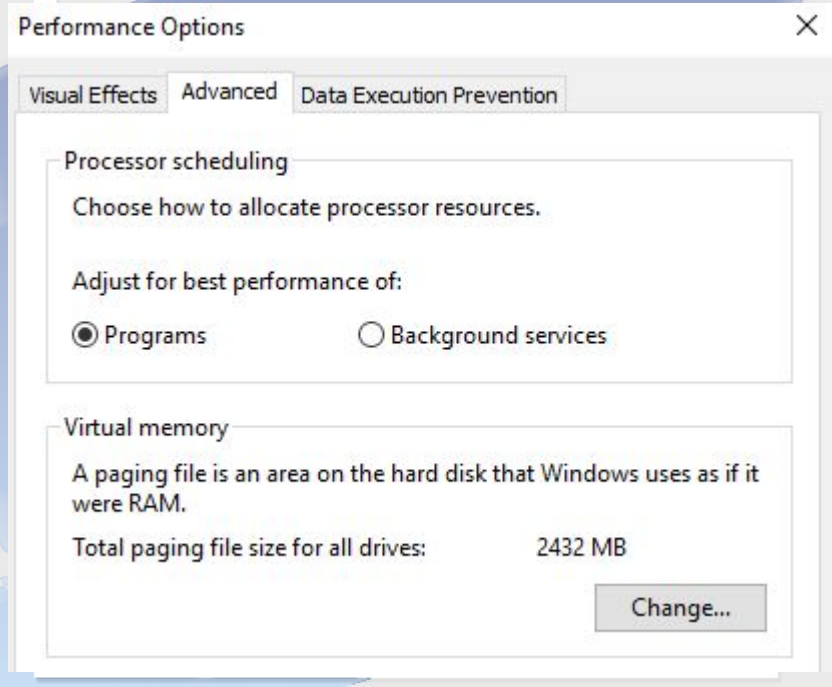
kerchoo



Perf - Adv

Remember LSP's "Clear virtual memory at shutdown"?

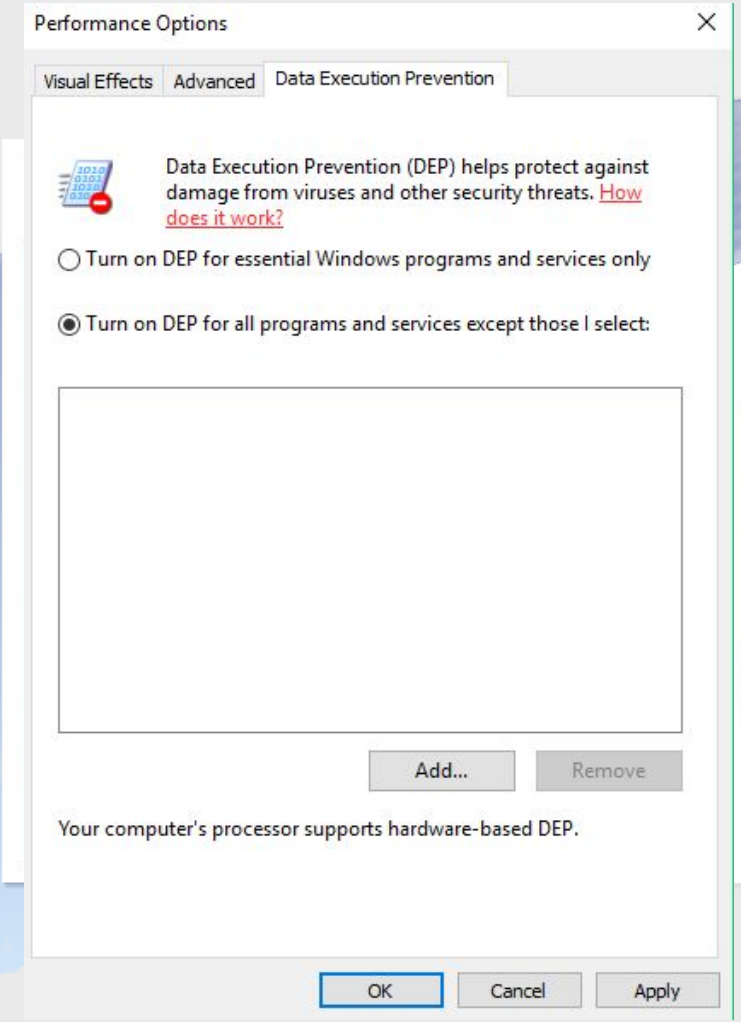
Here it is again



DEP

Performs additional checks on memory to prevent malicious code from running

Should turn on for ALL

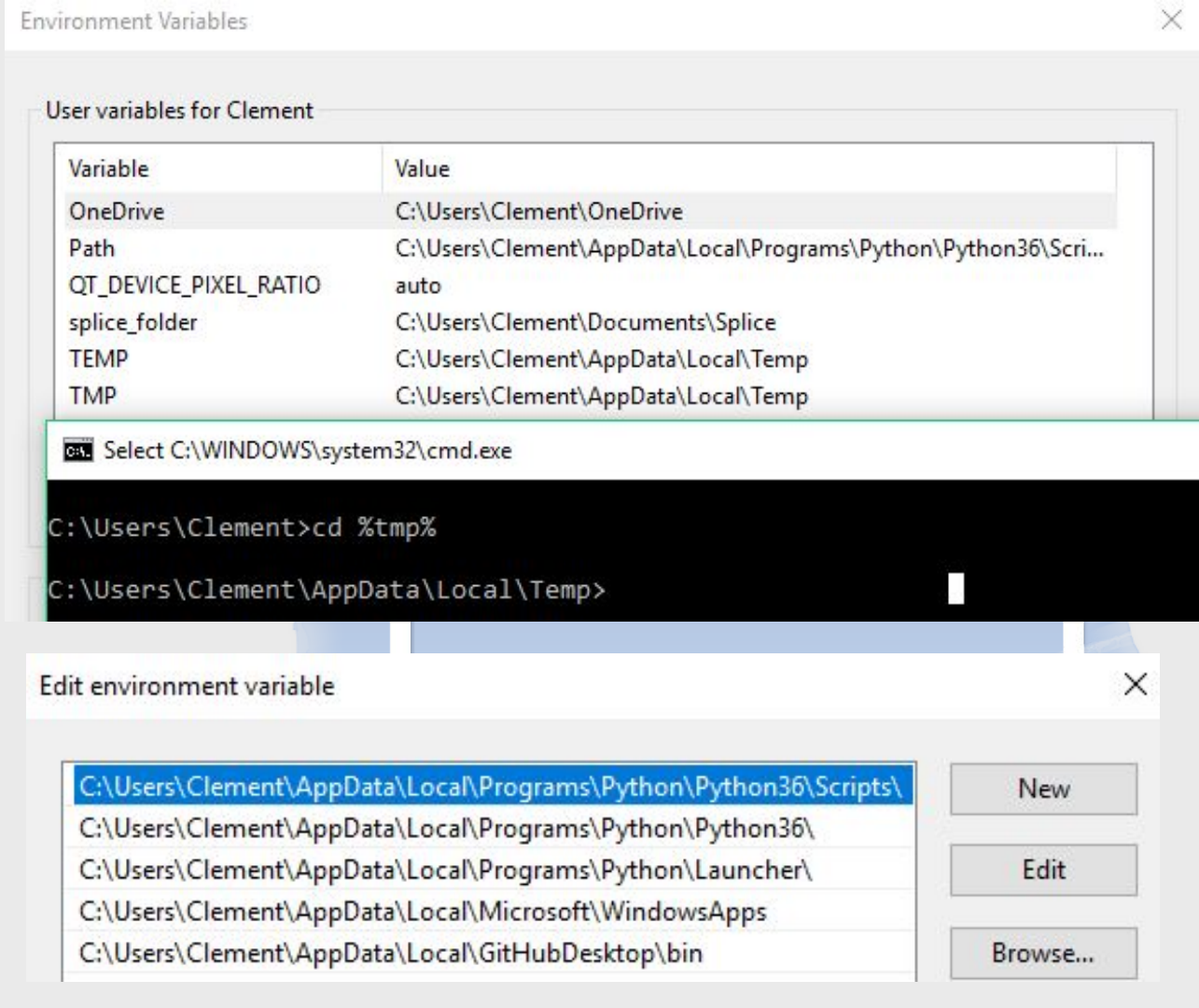


Env vars

Path is a special variable:

- Specifies directories to look for executables

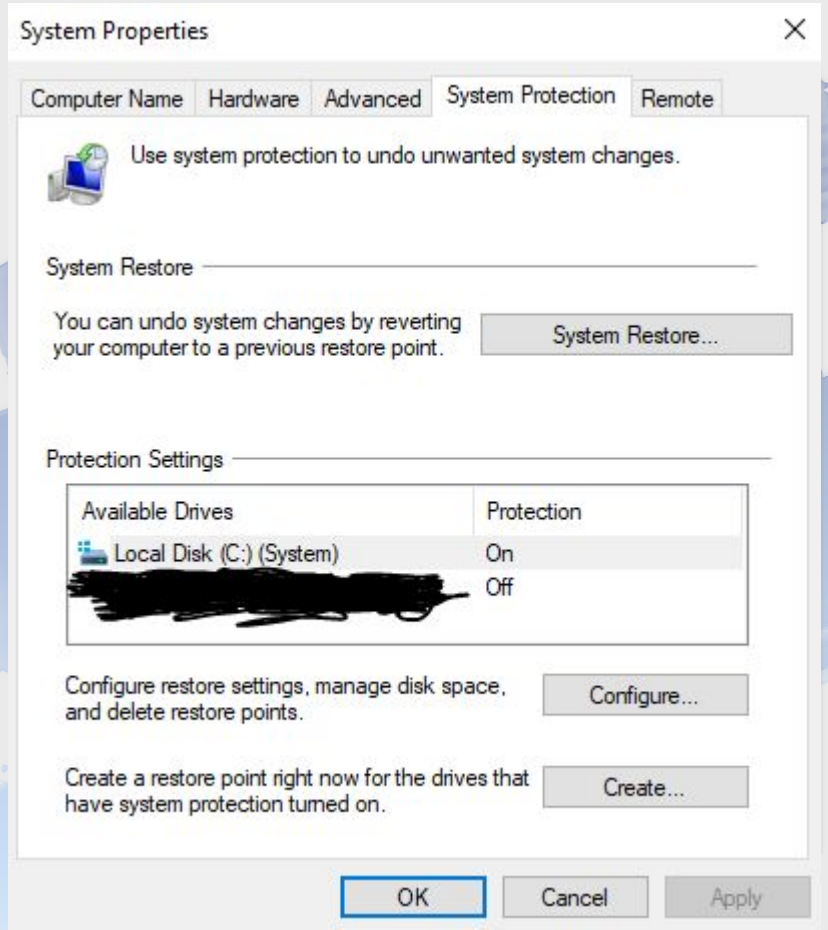
Take a look at Path above



System Protection

Set up restore point in case of failure

Important in the real world



Remote

You can also modify which users can Remote Desktop

As a review, best security practices are?

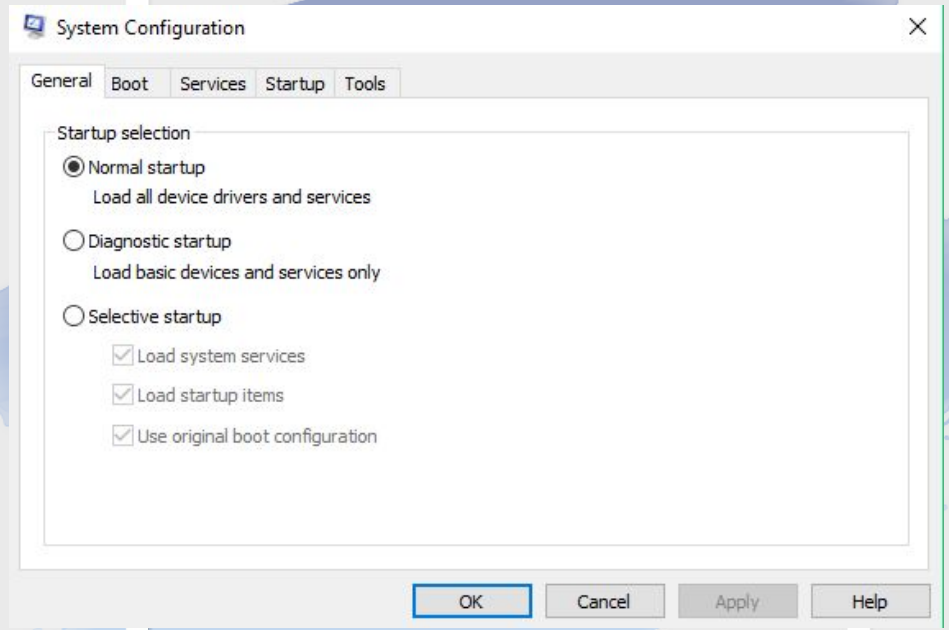
- No remote assistance
- Remote Desktop disabled

5.2 System Configuration Utility

System Conf. Utility

Run: “msconfig”

For startup configuration

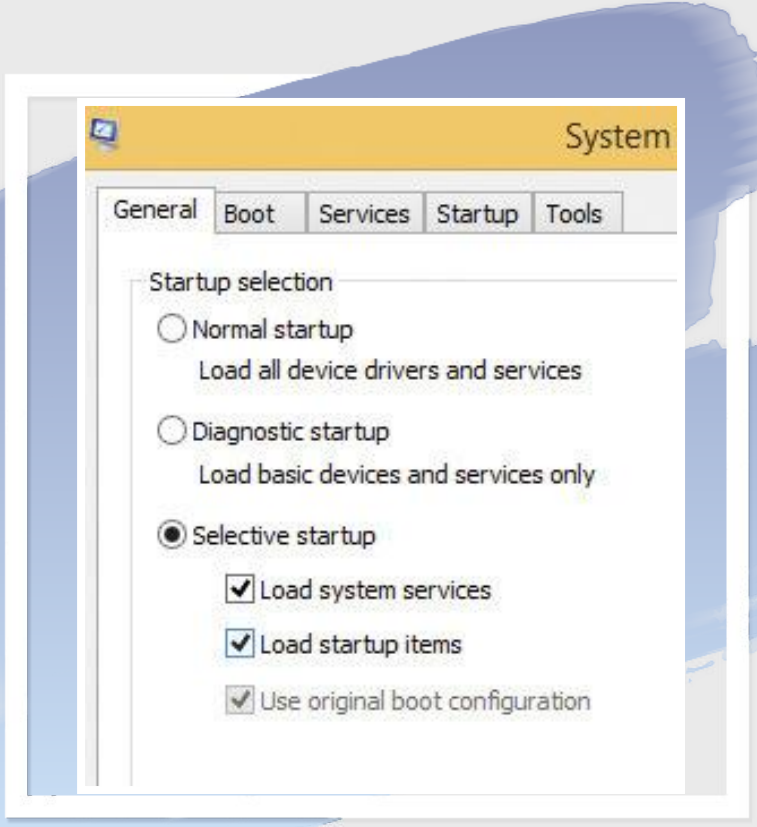


General

Choose specific startup type

- Normal
- Diagnostic
  - Only basic drivers and services start
- Selective
  - Choose what loads

Good for startup troubleshooting

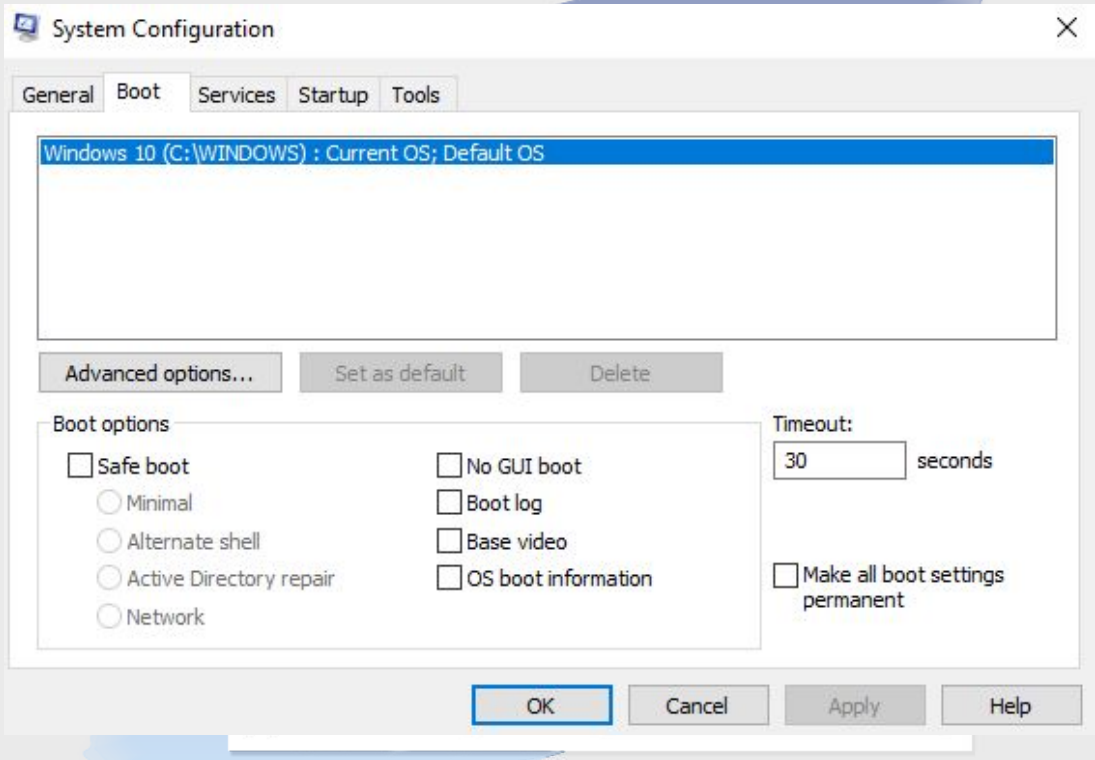


Boot

Control/modify Windows boot environment

Safe boot includes many options

Only really useful for boot troubleshooting

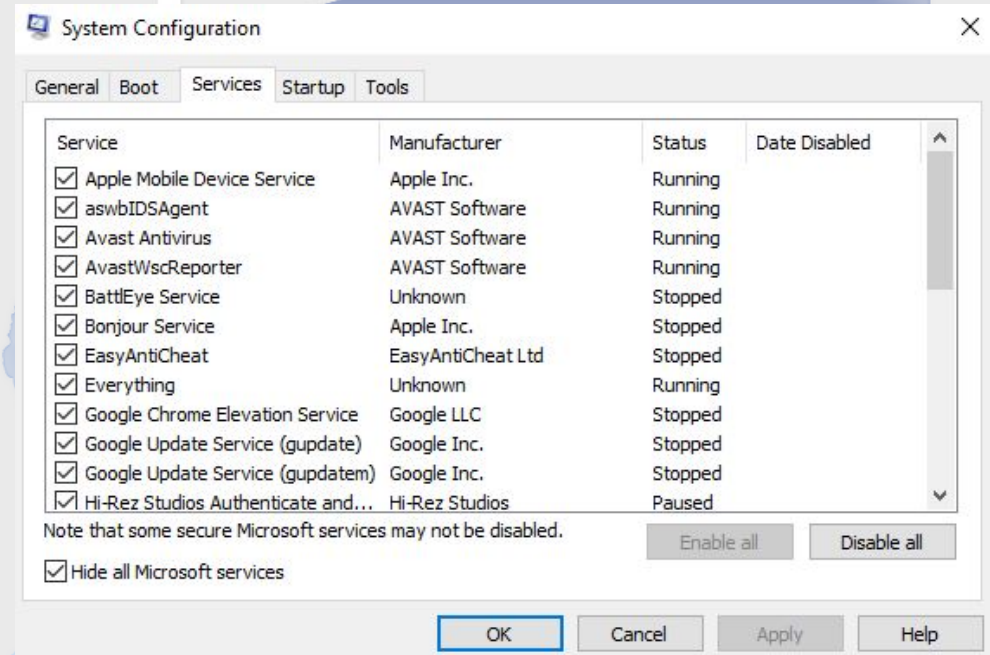


Services

Select which services are started at boot

Most useful thing is “Hide all Microsoft services”

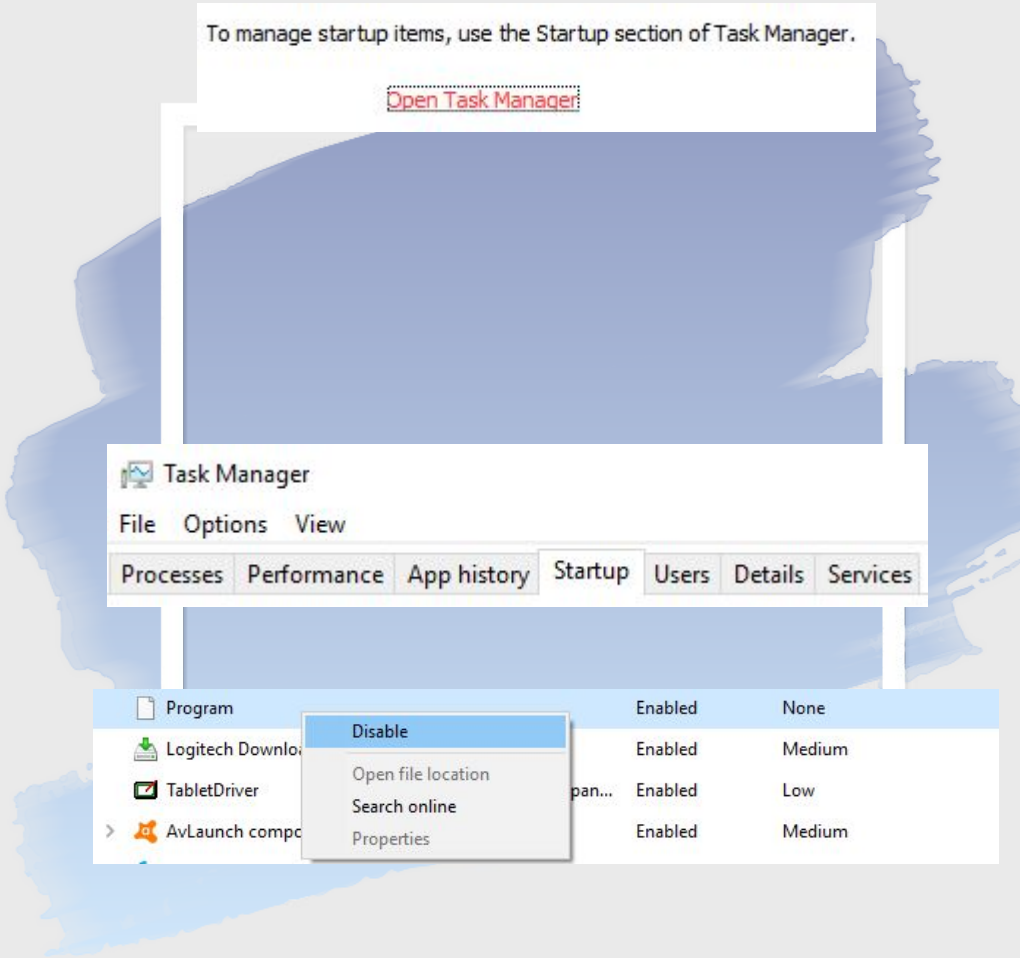
- Shows you custom services



Startup

Redirects you to Task Manager

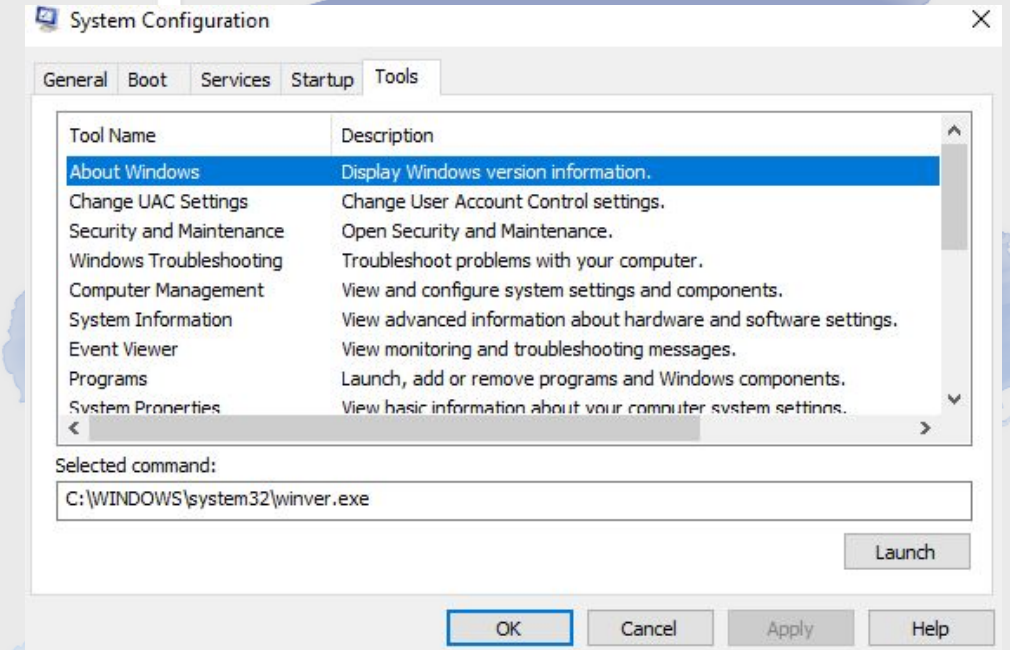
Make sure nothing bad is running at startup



Tools

Quick access to many different tools available from Windows

Not much new here, just a center point for many tools you've seen before



5.3 Monitoring Tools

Monitoring

Important in realizing on-off or repeated suspicious activity

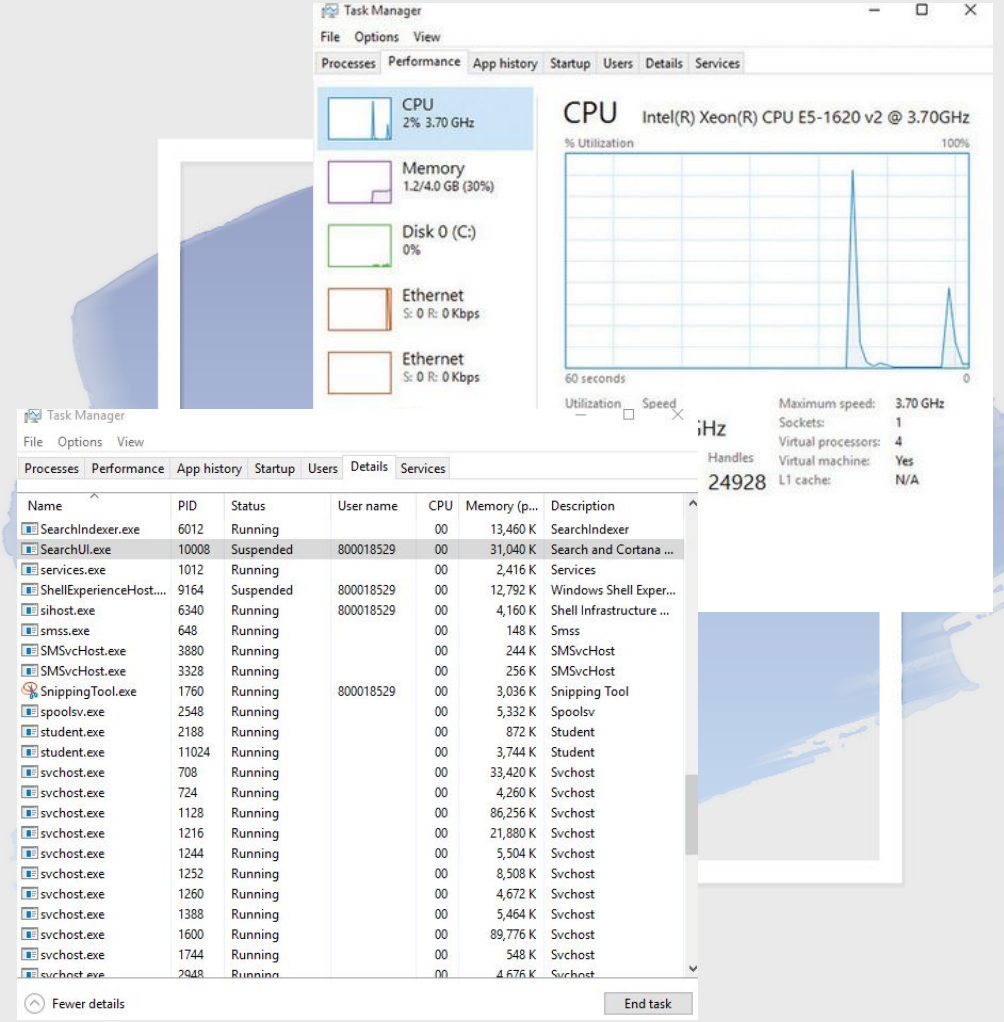
Really helpful when being attacked by hackers

Task Manager

Quick access by
right-clicking taskbar

Performance spikes can be
indicative of something

Details tab shows process
names

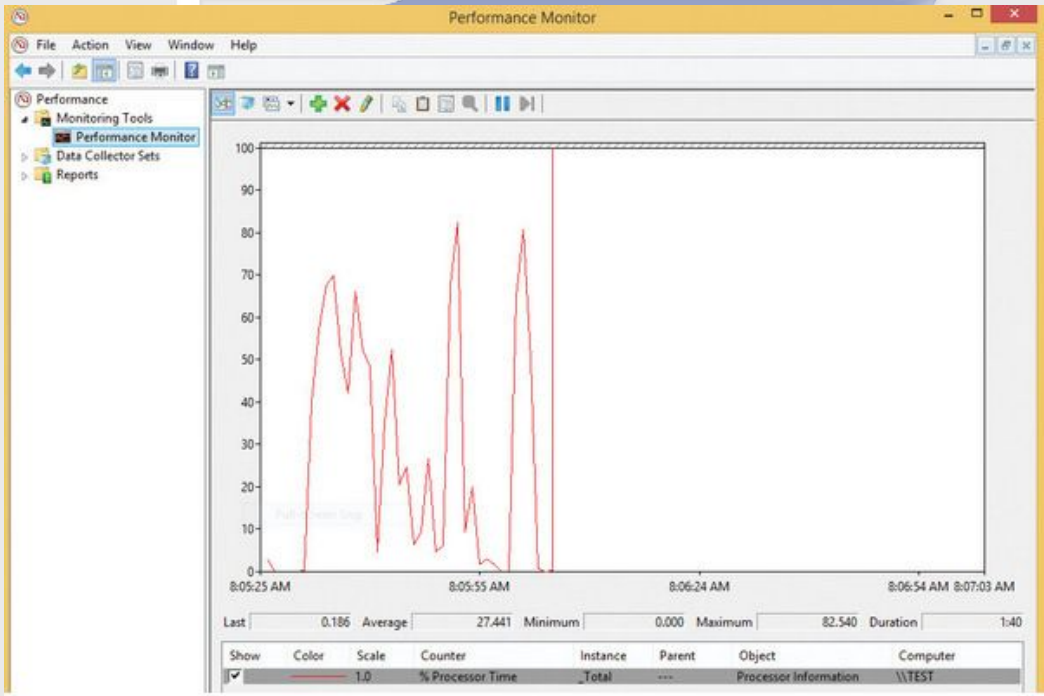


Performance Monitor

Good for specific measurements

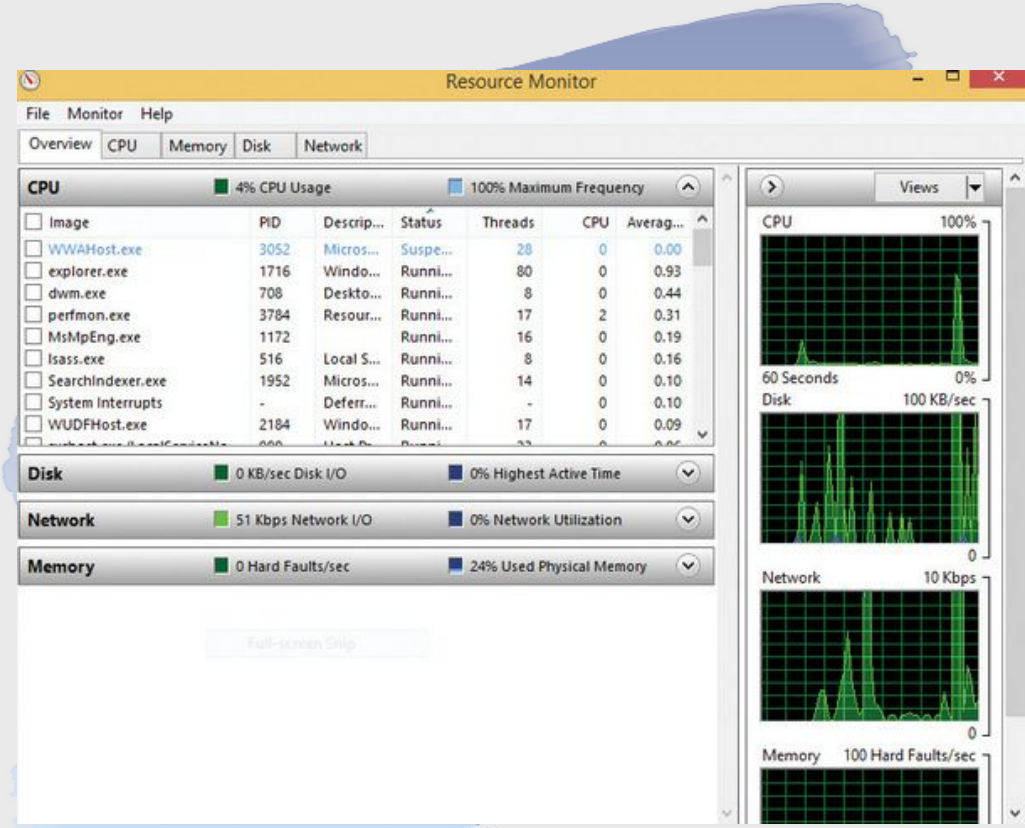
Counter - specific measurement for objects

- Counters exist for cache, memory, paging file, disk, etc.



Resource Monitor

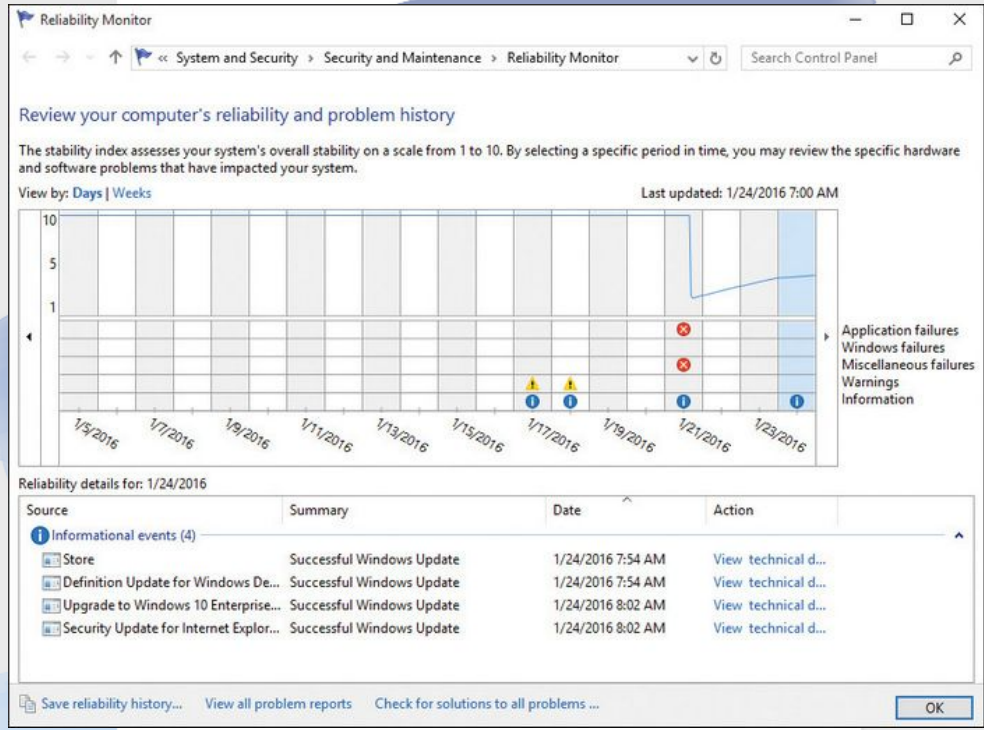
Like task manager, but splits between CPU, Disk, Network, and Memory for each process



Reliability Monitor

Most useful for servers or other critical computers that need to provide reliable service

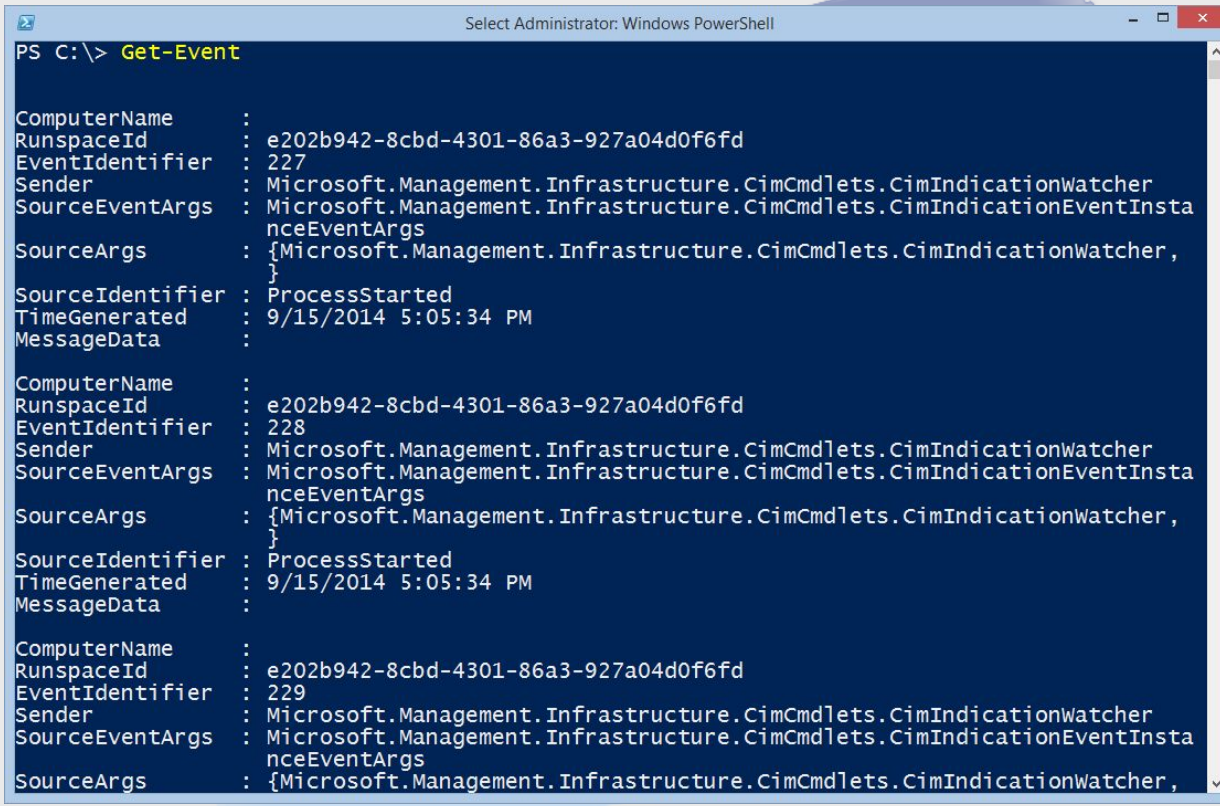
Quickly view critical reliability issues like failed updates, crashes, etc.



Customized monitoring via terminal

Use customized batch/powershell script

Can customize to show only new processes, processes in Windows dir, etc.



Summary of Monitors

- Task Manager
- Performance Monitor
- Resource Monitor
- Reliability Monitor
- Customized Terminal Monitor

Task Manager and Customized Terminal Monitor are arguably the best to use in cybersecurity

Others are more for tech stats/support

5.4 Windows Registry

Registry

The complicated heart of all Windows software and hardware

Include:

- Port configurations
- App preferences
- User profiles
  - Specific conf for each user

Loaded into RAM during boot, updates continuously

Reg Terminology

regedit - tool to edit registry

hive - branch of registry for organization

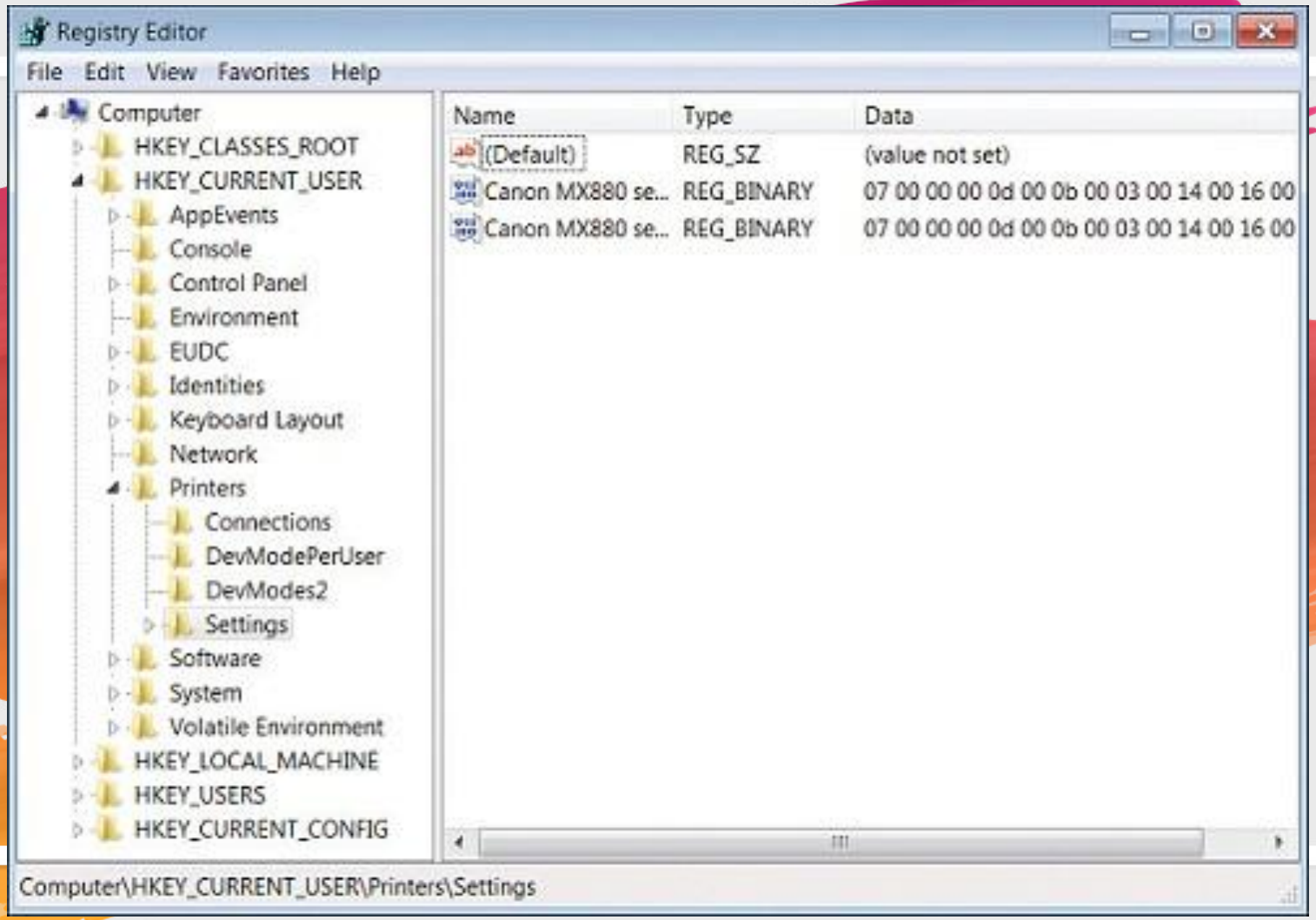
key - like a folder, contains values

value - registry setting

data - the state of a setting

type - the type of the data

can be SZ for string, BINARY, DWORD, MULTI_SZ, QWORD


Registry Editor
File Edit View Favorites Help
Computer
HKEY_CLASSES_ROOT
HKEY_CURRENT_USER
AppEvents
Console
Control Panel
Environment
EUDC
Identities
Keyboard Layout
Network
Printers
Connections
DevModePerUser
DevModes2
Settings
Software
System
Volatile Environment
HKEY_LOCAL_MACHINE
HKEY_USERS
HKEY_CURRENT_CONFIG
Name
Type
Data
(Default)
REG_SZ
(value not set)
Canon MX880 se...
REG_BINARY
07 00 00 00 0d 00 0b 00 03 00 14 00 16 00
Canon MX880 se...
REG_BINARY
07 00 00 00 0d 00 0b 00 03 00 14 00 16 00
Computer\HKEY_CURRENT_USER\Printers\Settings

Five subtrees

All hives begin with HKEY

Registry subtree	Subtree function
HKEY_CLASSES_ROOT	• Holds file associations and file links • Allows the correct application to start when you double-click a filename
HKEY_CURRENT_USER	• User configuration, such as software settings, desktop appearance, user-created folders
HKEY_LOCAL_MACHINE	• Global hardware configuration • List of hardware components with their software drivers • Settings for each device
HKEY_USERS	• User preferences
HKEY_CURRENT_CONFIG	• Hardware profile info used on boot

Where is GPEDIT?

GPEDIT settings are split into

- Computer Configuration
- User Configuration

Corresponding subtrees are:

- LOCAL_MACHINE
- CURRENT_USER

Making changes

Relatively simple

1. Search for registry value
2. If value doesn't exist, create a new one
3. Set the data

It's a good idea to backup before changing something you don't know

File > Export

Saves as a .reg file

Why bother with registry?

Sometimes malicious settings are set in critical areas that wouldn't normally be found with GPEDIT or other tools

Typically registry searching is saved for last, as it is the most obscure

END of Part I
Any questions?